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Cryptography Theory and Practice

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Question 1

a. The BPV algorithm works in an online/offline structure where the offline phase, with inputs N and e where N=p*q, precomputes n random values and their modular exponent stored as (x, x^e mod N). Then another security parameter T is decided, based on subset size k, each exponentiation chooses T of these pairs and multiplies the x^e mod N values. This results in every modular exponentiation only requiring T multiplications instead of 1.5L in a square-multiply method, meaning the number of multiplications needed is constant and not tied to the length of the exponent.
b. Given L=4096 and K=16 then the estimated number of multiplications needed is T=L/K=4096/16=256. This means that, given the number of multiplications for square-multiply is 1.5L, the square-multiply needs 6,144 multiplications, resulting in 5,888 multiplications saved.

Question 2

a. Since the device can store at most 127 precomputed values and for a given input K 2^K - 1 precomputed values are needed, K=7 is the largest value possible without exceeding the device's memory limit.
b. Given the approximate number of multiplications needed is T=L/K with an L=1024 and K=7, the approximate multiplications needed is 146.
c. With 146 multiplications needed for the BPV method and 1536 multiplications needed for square-multiply, 1.5*1024=1536, there is a 91.5% reduction in the number of multiplications needed.

Question 3

a. Since the standard square-and-multiply algorithm requires around 1.5L multiplications are required for each exponentiation, a system with 2048-bit exponents will require 3072 multiplications.
b. For the BPV offline phase, the approximate number of multiplications required $2^k - 1$. With K=8, the approximate amount is 255.
c. The total number of BPV multiplications is 511.
d. The BPV algorithm achieves approximately 84% in computational savings.

Question 4

a. Considering $T = \frac{L}{K}, L = 3072$, and a strict maximum of 300 multiplications for the VPN online phase, the largest K value is 11.
b. With L=3072 and K=11, there would be $T = \frac{L}{K} = \frac{3072}{11} = 280\ multiplications$
c. K controls the amount of work done offline vs online. With a larger K, there is less online work ($T = \frac{L}{K}$) but more stored values ($2^k - 1$) $and\ preparation$. As K increases, there are fewer groups to process online but more possible values that need to be precomputed and stored offline. The tradeoff for K is therefore finding the right size to balance preparation and memory vs online computations, which will be dictated by the environment, hardware limitations and application.

[1] Boyko, Victor et al. "Speeding up Discrete Log and Factoring Based Schemes via Precomputations." *International Conference on the Theory and Application of Cryptographic Techniques* (1998).